Electric forces and fields

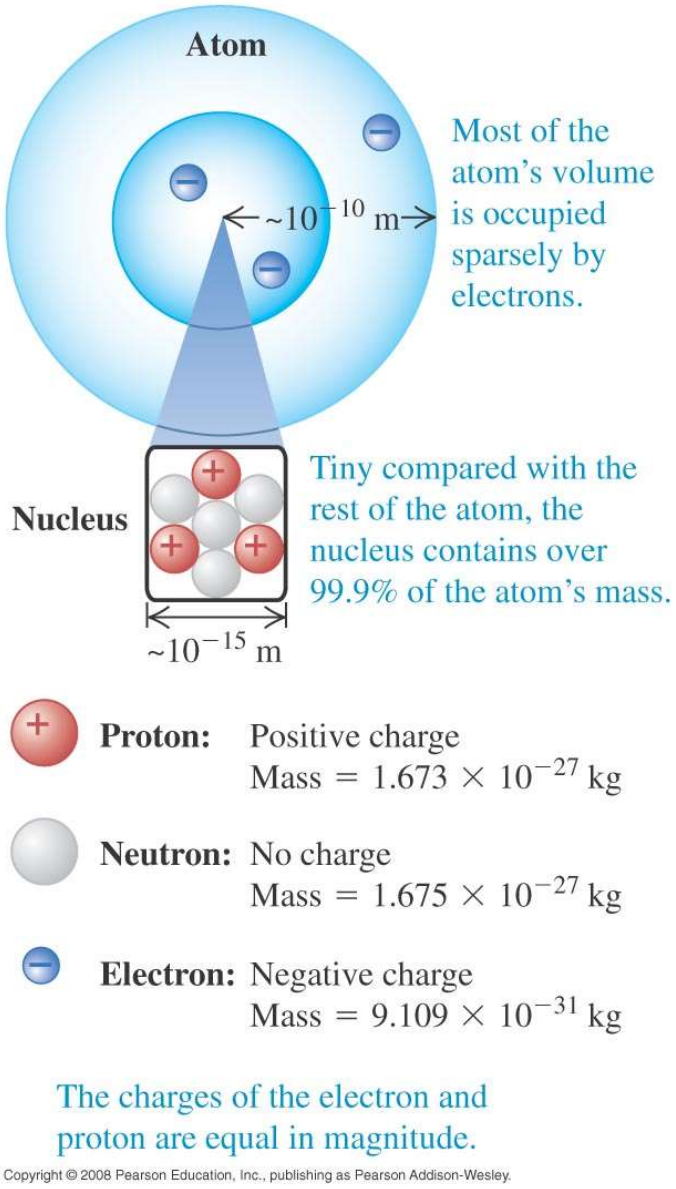
Atomic structure

- Atoms are made up of :
  - negative electrons
  - positive protons
  - neutrons

- There are two kinds of charges and we differentiate by assigning “+” and “-” signs.

- An electron has a charge of -1.6x10^{-19} C.

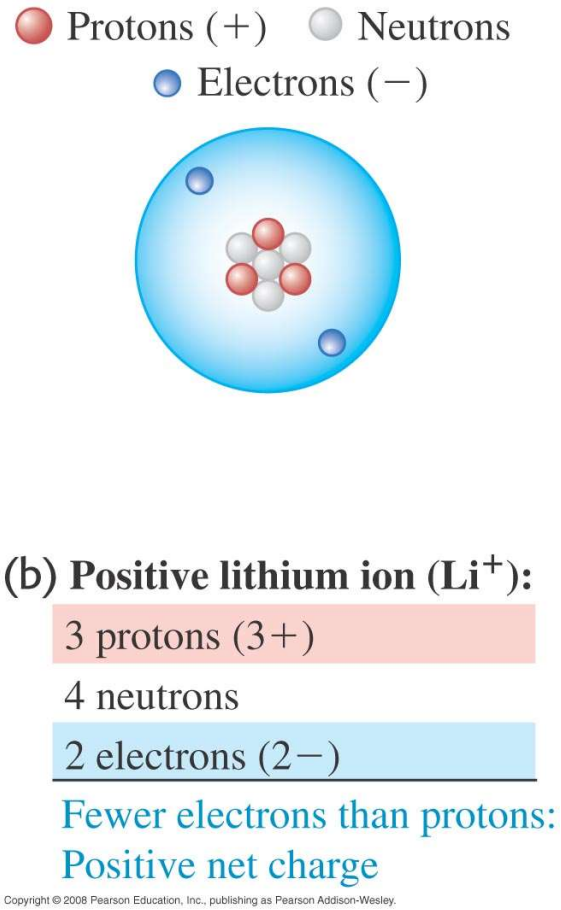
- A proton has a charge of +1.6x10^{-19} C.



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Ions and isotopes

- An ion is an atom with missing or extra electrons.
- A positive lithium ion has 3 protons and 2 electrons.
- A negative lithium ion has 3 protons and 4 electrons.
- Isotopes are atoms of the same element having the same number of protons but different number of neutrons.



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Example 1

How many electrons will give you a total charge of -32×10^{-6} C?

Coulomb's Law

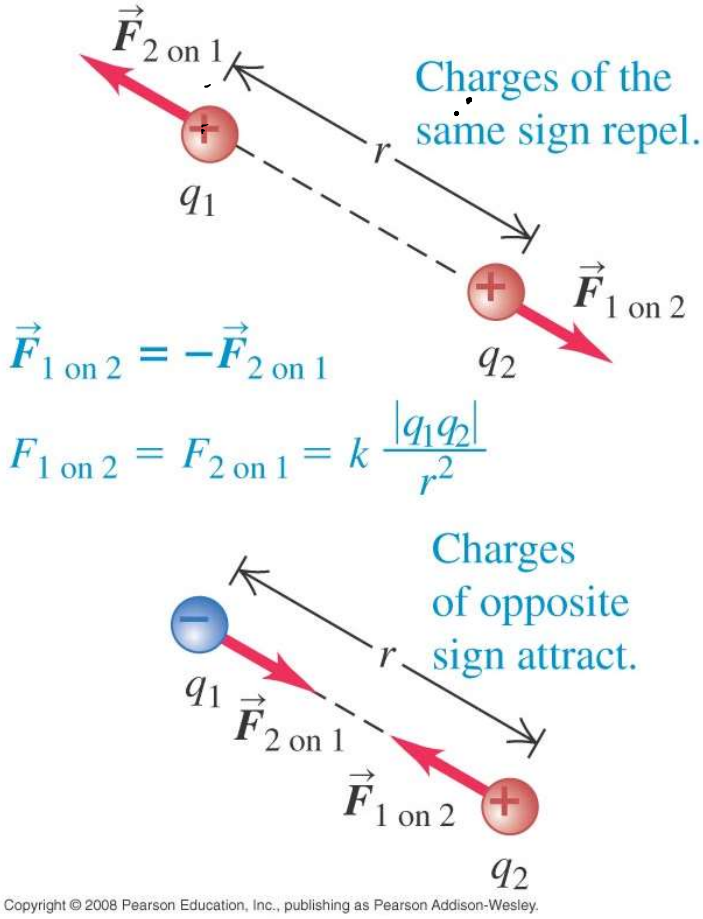
- Force between two stationary point charges is given by **Coulomb's Law**,

$$\vec{F} = k\frac{q_1 q_2}{r^2}\hat{r}$$

- The magnitude of the force is

$$F = k\frac{|q_1 q_2|}{r^2}$$

where k = 8.9876×10^9 Nm^2/ C^2 is the Coulomb constant.



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Permittivity

- The Coulomb's constant can also be written as :

$$k = \frac{1}{4\pi\varepsilon_o}$$

where ε_o is called the permittivity of free space (vacuum) and has a value of 8.854×10^{-12} $C^2/N\,m^2$.

- If the charges are in some medium, then $k = \frac{1}{4\pi\varepsilon}$

where ε is the permittivity of the medium.

Relative permittivity

- The ratio $\varepsilon_r = \frac{\varepsilon}{\varepsilon_o}$

  is known as relative permittivity.

- When sodium chloride is put into water of relative permittivity 80, the force between a sodium and chloride ion is reduced by 80 times.

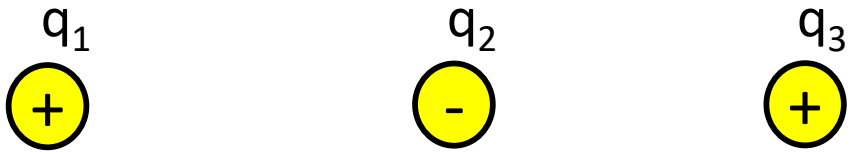
- This explains the high solubility of common salt (NaCl) in water.

Superposition of forces

- The superposition principle states that the resultant force on any one charge equals the vector sum of the forces exerted by the other individual charges that are present.

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \ldots$$

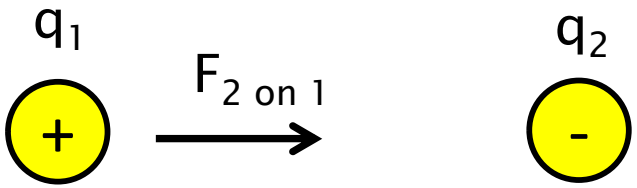
Force on a charge due to other charges



- Imagine only charge 3 is present and draw the force on charge 1 due to charge 3.



- Next imagine only charge 2 is present and draw the force on charge 1 due to charge 2.



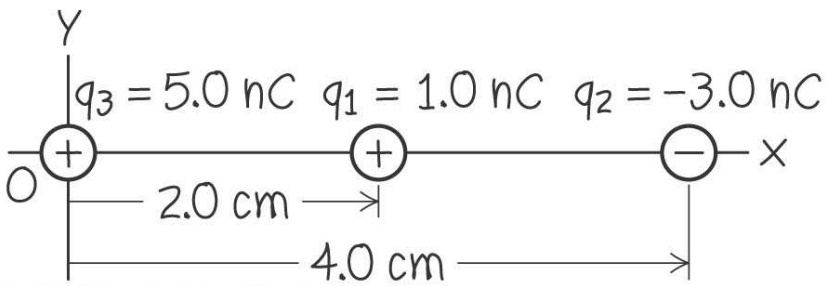
- The force on charge 1 is the **vector sum** of the two forces.



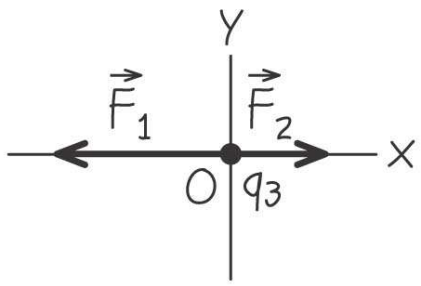
Example 2

Two point charges are located on the x-axis of a coordinate system: $q_1 = 1.0$ nC is at x = +2.0 cm and $q_2 = -3.0$ nC is at x = +4.0 cm. What is the total electric force exerted by q_1 and q_2 on a charge $q_3 = 5.0$ nC at x = 0?

(a) Our diagram of the situation

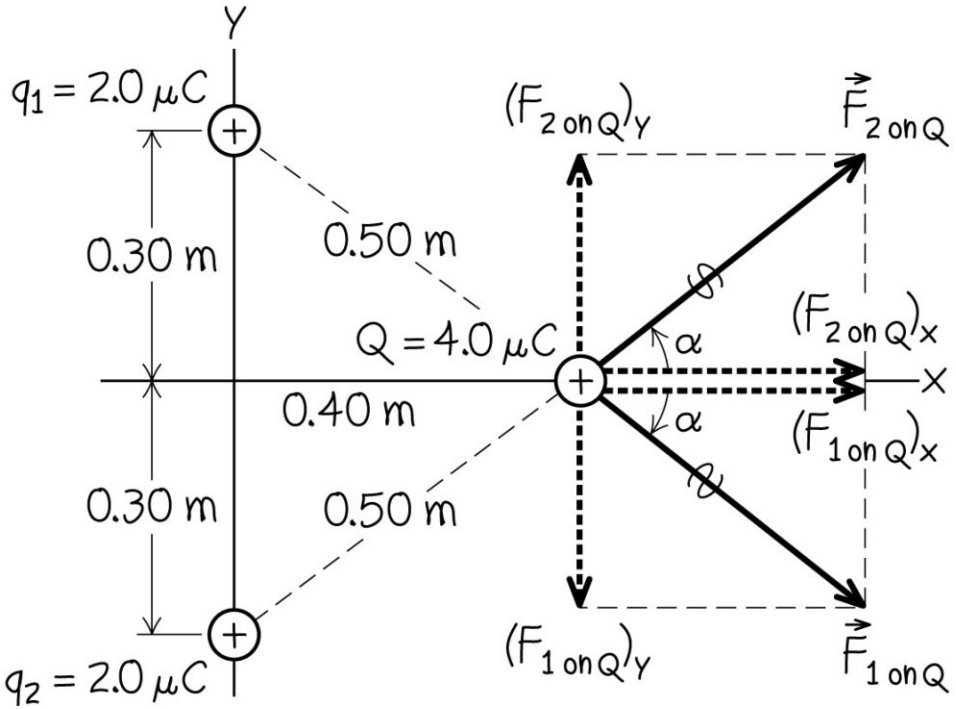


(b) Free-body diagram for q_3



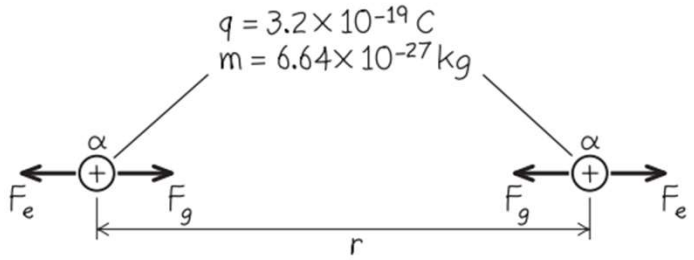
Example 3

Two equal positive charges point charges $q_1 = q_2 = 2.0\ \mu C$ are located at $x = 0$, $y = 0.30$ m and $x = 0$, $y = -0.30$ m respectively. What are the magnitude and direction of the total electric force that q_1 and q_2 exert on a third charge $Q = 4.0\ \mu C$ located at $x = 0.4$ m, $y = 0$?



Comparison of Electric force and Gravitational force

An α particle (the nucleus of a helium atom) has mass $m = 6.64 \times 10^{-27}$ kg and charge $q = +2e = 3.2 \times 10^{-19}$ C. Compare the magnitude of the electric repulsion between two α ("alpha") particles with that of the gravitational attraction between them.



$$F_e = \frac{1}{4\pi\epsilon_0}\frac{q^2}{r^2} \qquad F_g = G\frac{m^2}{r^2}$$

These are both inverse-square forces, so the r^2 factors cancel when we take the ratio:

$$\frac{F_e}{F_g} = \frac{1}{4\pi\epsilon_0 G}\frac{q^2}{m^2}$$

$$= \frac{9.0 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2}{6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2}\frac{(3.2 \times 10^{-19} \text{ C})^2}{(6.64 \times 10^{-27} \text{ kg})^2}$$

$$= 3.1 \times 10^{35}$$

This astonishingly large number shows that the gravitational force in this situation is completely negligible in comparison to the electric force. This is always true for interactions of atomic and subnuclear particles. But within objects the size of a person or a planet, the positive and negative charges are nearly equal in magnitude, and the net electric force is usually much *smaller* than the gravitational force.

Key

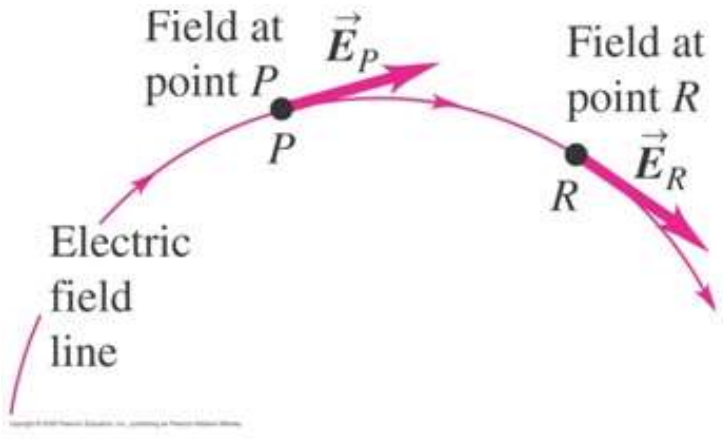
Key words in a problem

1. When → Calculate time
2. Initial → at $t = 0$
3. Rest → velocity $= 0$
4. Where → position / distance
5. change in x → $\Delta x = x_f - x_i$ (initial value, final value)
6. Compare → ratio : A/B
7. Difference → $x_A - x_B$

Show or Prove $\Rightarrow$ LHS. = RHS.

Electric field

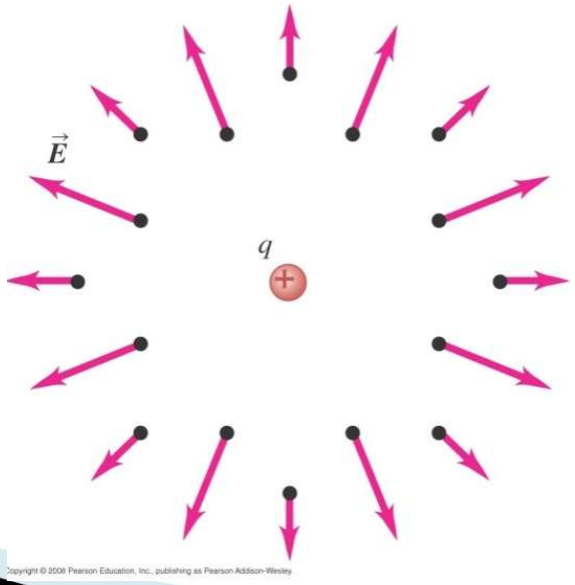
- An **electric field** is a region where an electric charge experiences an electric force.
- Electric fields are represented by field lines.
- The tangent at any point on a field line is the direction of the force that a positive charge would experience when placed at that point.



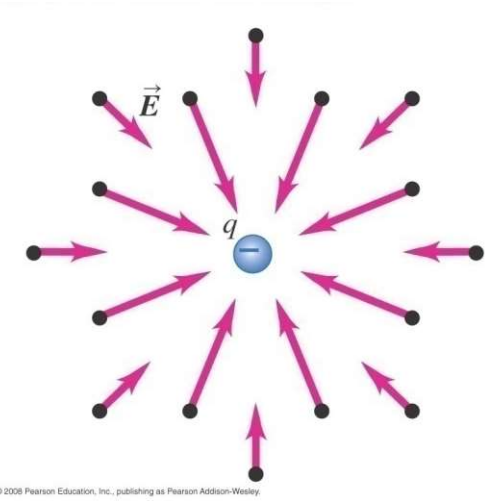
Electric field lines of point charges

- For a single isolated positive point charge, the field lines always go away from the positive charge.

- For a single isolated negative point charge, the field lines always go towards the negative charge.



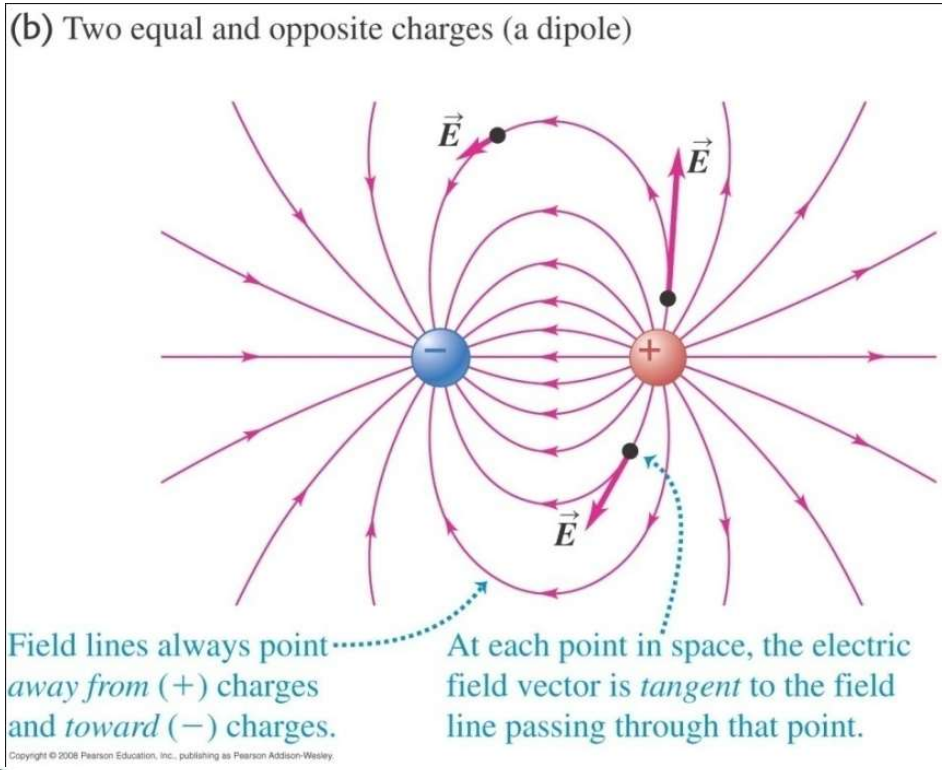
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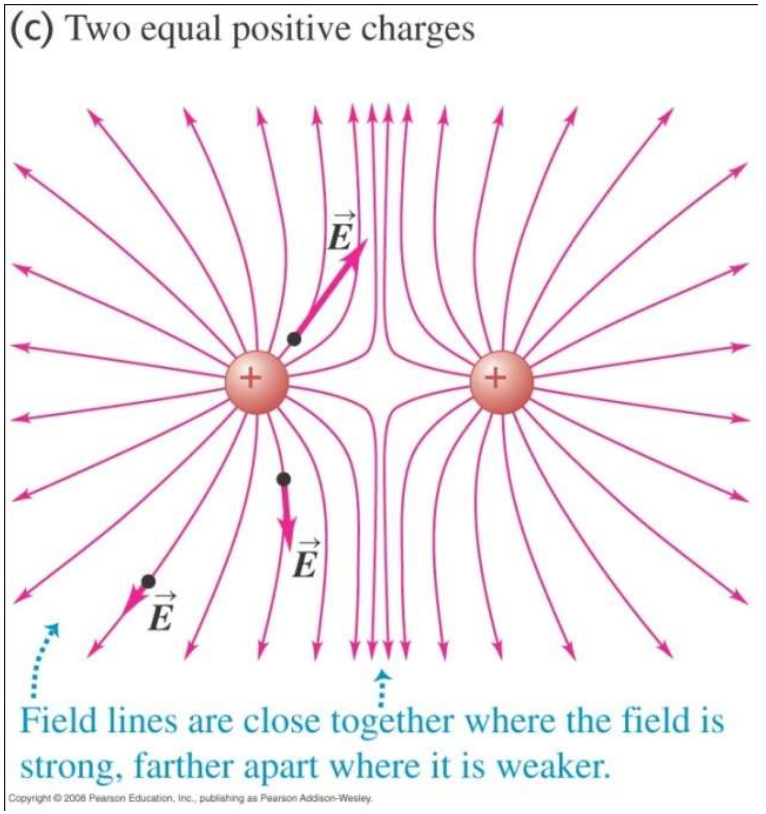


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Electric field lines of two nearby point charges

- The electric field of two nearby point charges is the resultant of the fields of the two charges.





Calculating electric field of a point charge

- From Coulomb's Law, force on q_0 is

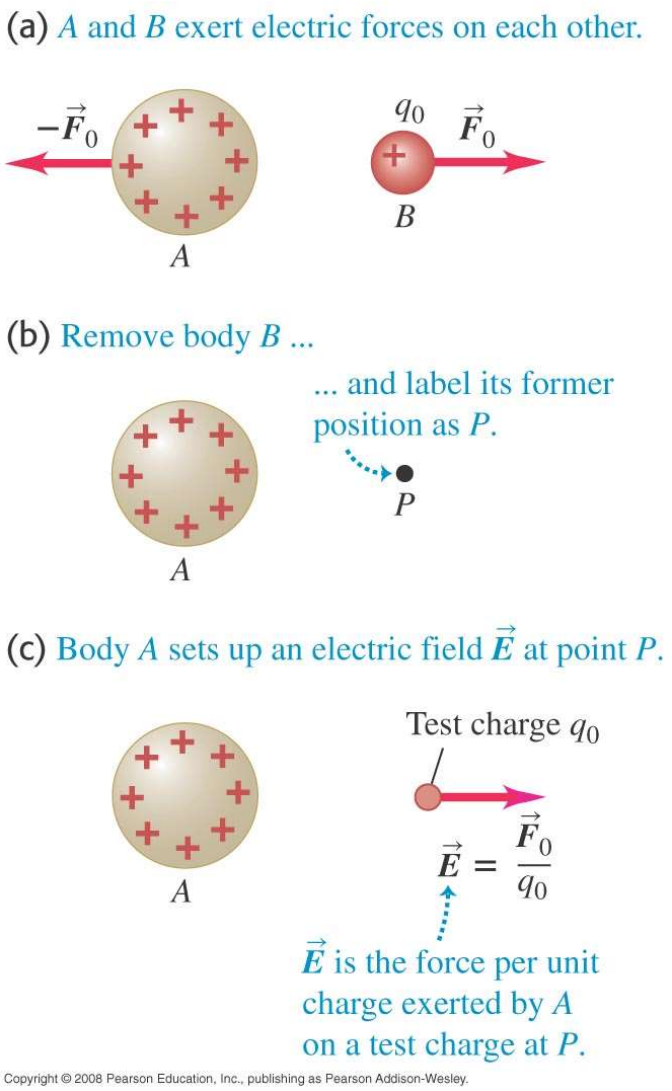
$$\vec{F}_0 = k\frac{q_A q_0}{r^2}\hat{r}$$

- The electric force per unit charge on the q_0 due to q_A is defined as

$$\vec{E} = \frac{\vec{F}_0}{|q_0|} = k\frac{q_A}{r^2}\hat{r} \quad \text{point charge only}$$

- The quantity $\vec{E}$ is the electric field at the position P.

- Its SI unit is N/C.



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A point charge in an electric field

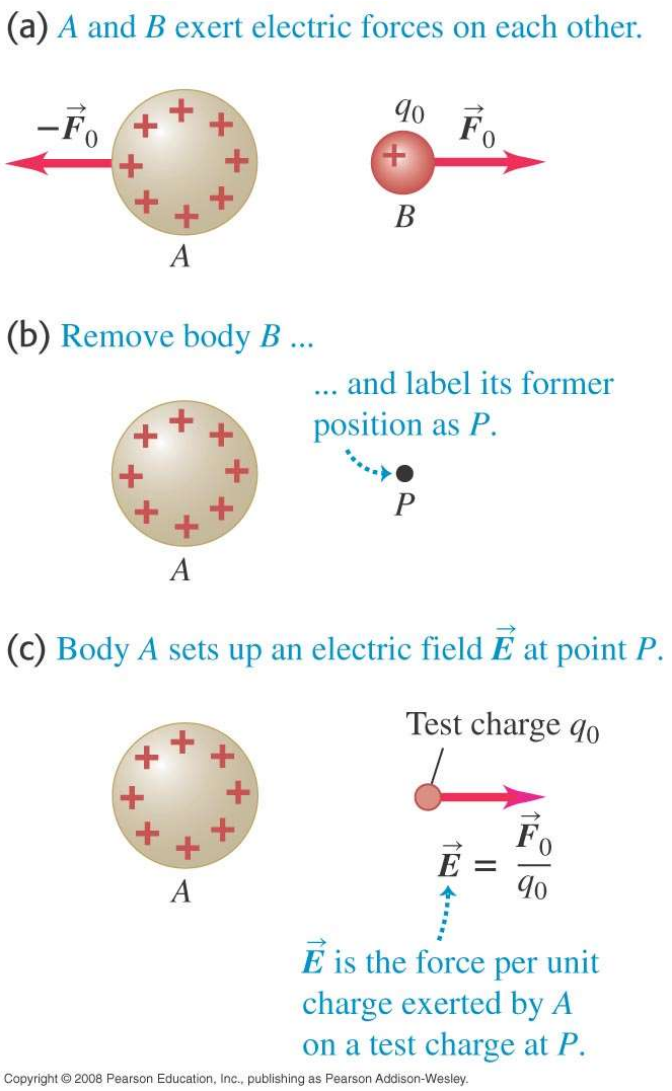
- The force on q_0 is therefore

$$\vec{F} = q_0\vec{E}$$

- Note that the electric field E is due to q_A (and not due to q_0 itself!).

- For a positive charge, Electric field, ***E*** and Force, ***F*** are in the same direction, while for a negative charge, they are in opposite direction to each other.

- Its SI unit is N/C.

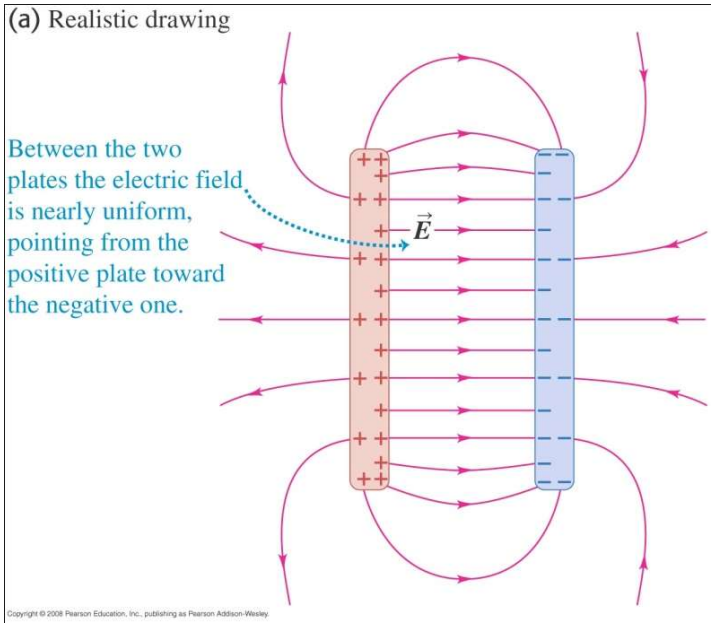


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Electric field between two parallel plates

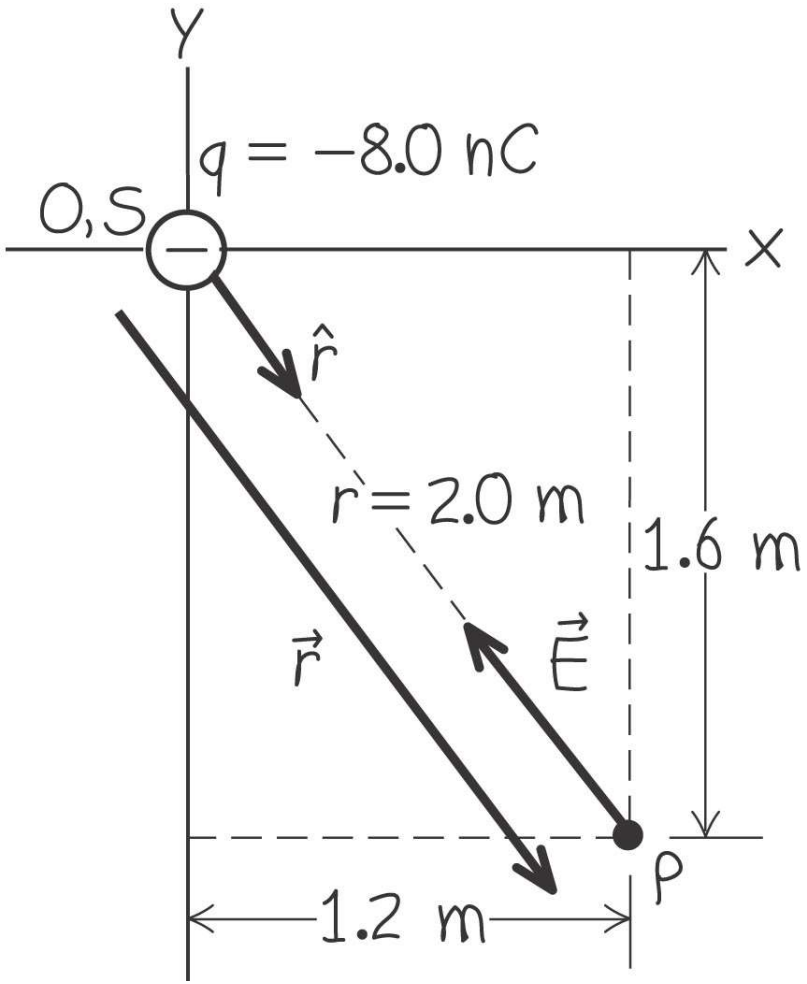
- The electric field is uniform except near the edges of the plates.

- A charge within the uniform field will experience the same force no matter where it is placed.



Example 4

A point charge q = −8.0 nC is located at the origin. Find the electric field vector at the point x = 1.2 m, y = − 1.6 m.



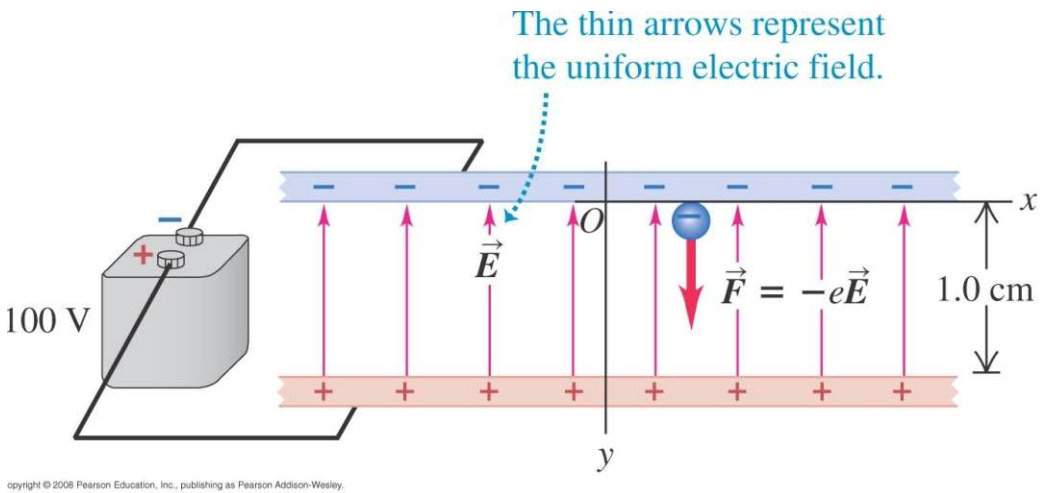
Example 5

The diagram shows a uniform electric field of magnitude E = 1.00 x 10^4 N/C. Mass of an electron is 9.11 × 10^{-31} kg.

a) If an electron is released from rest at the upper plate, what is its acceleration?

b) What speed and kinetic energy does the electron acquire travelling 1.0 cm to the lower plate?

c) How much time is required for it to travel this distance?



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Superposition of electric fields

- If we have a distribution of charge, we can imagine it to be made up of many point charges.

- The total force on a charge q_0 due to the fields of these point charges is

$$\vec{F}_0 = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \ldots = q_0\vec{E}_1 + q_0\vec{E}_2 + q_0\vec{E}_3 + \ldots$$

- Hence, the overall field is the superposition of electric fields which is

$$\vec{E} = \frac{\vec{F}_0}{q_0} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \ldots$$

Superposition of electric fields

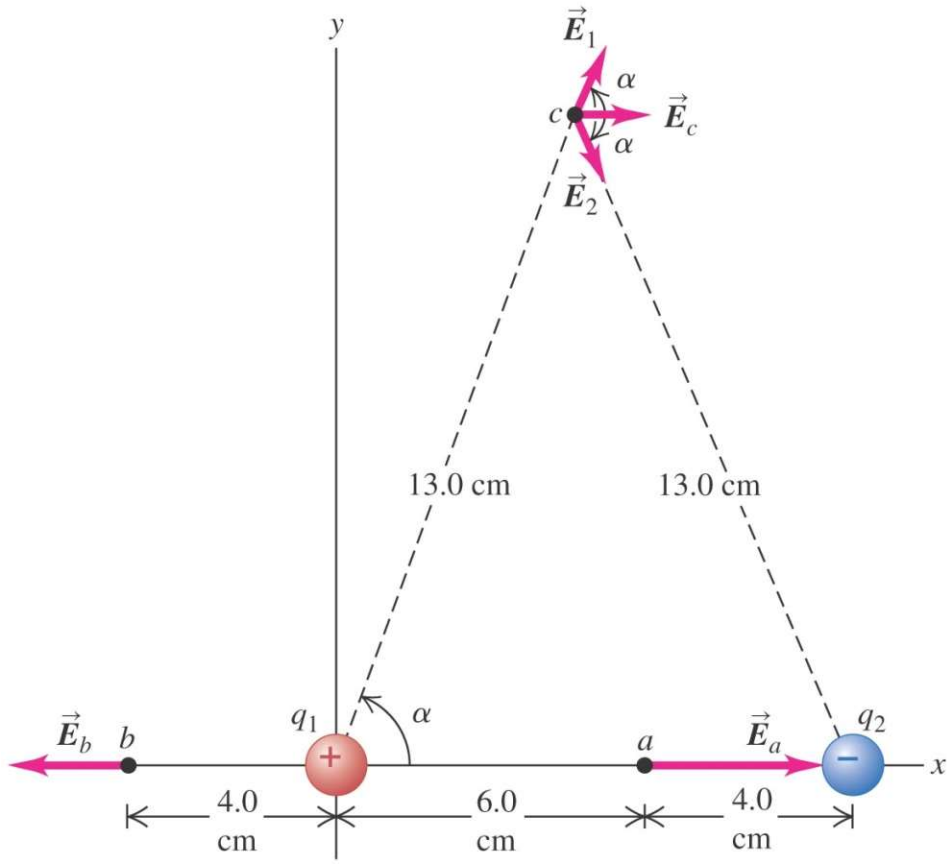
When there are more than one charges, electric field at point P due all the charges are vector sum of field due to each charge, just like superposition of forces:

$$\vec{E} = k_e \left(\frac{q_1}{r_1^2}\hat{r}_1 + \frac{q_2}{r_2^2}\hat{r}_2 + \frac{q_3}{r_3^2}\hat{r}_3 + \cdots + \frac{q_N}{r_N^2}\hat{r}_N \right) = k_e \sum_{i=1}^{N} \frac{q_i}{r_i^2}\hat{r}_i,$$

where r_i is the distance of charge q_i from point P and $\hat{r}_i$ is the unit vector directed from q_i toward P.

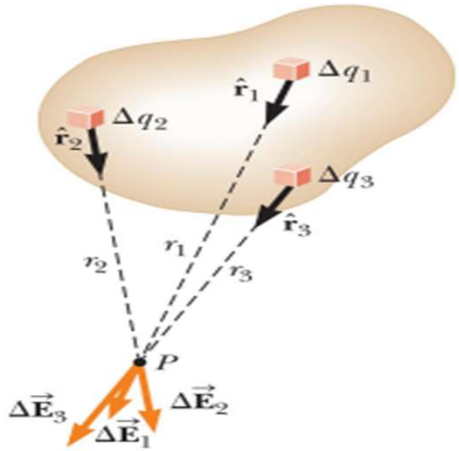
Example 6

Point charges, q_1 = +12 nC and q_2 = −12 nC, placed 10 cm apart (see figure) . Compute the total electric field at (a) point a, (b) point b, (c) point c.



Electric fields for continuous charge distributions

Figure 1: The electric field at P due to a continuous charge distribution is the vector sum of the fields $\Delta\vec{E}_i$ due to elements Δq_i. Three sample elements are shown. (Courtesy Serway & Jewett).



where ρ has units of units of coulombs per cubic meter (C/m^3).

For continuous charge distributions, electric field at a point P is the vector sum of the fields $\Delta\vec{E}_i$ due to all the elements Δq_i of the charge distribution. Three sample elements are shown in the Fig.1 A limit of this sum when the number of such elements are infinite, we get an integral:

$$\vec{E} = \lim_{\substack{N\to\infty \\ \Delta q_i \to 0}} \sum_{i=1}^{N} k_e \frac{\Delta q_i}{r_i^2}\hat{r}_i = k_e \int \frac{dq}{r^2}\hat{r},$$

where r_i is the distance of the centroid C_i of the infinitesimal charge Δq_i from P and $\hat{r}_i$ is the unit vector directed from C_i to P.

Linear, Surface and Volume Charge Densities

If a charge Q is uniformly distributed along a line of length ℓ the linear charge density λ is defined by

$$\lambda = \frac{Q}{\ell},$$

where λ has units of units of coulombs per meter (C/m).

If a charge Q is uniformly distributed on a surface of area A, the surface charge density σ (Greek letter sigma) is defined by

$$\sigma = \frac{Q}{A},$$

where σ has units of units of coulombs per square meter (C/m^2).

If a charge Q is uniformly distributed throughout a volume V, the volume charge density ρ is defined as

$$\rho = \frac{Q}{V},$$

ρ has units of C/m^3.

Problem solving strategy to calculate Electric fields for continuous distribution of charges

$$\vec{E} = \lim_{\substack{N\to\infty \\ \Delta q_i \to 0}} \sum_{i=1}^{N} k_e \frac{\Delta q_i}{r_i^2}\hat{r}_i = k_e \int \frac{dq}{r^2}\hat{r},$$

If the charge is distributed (irrespective of whether it is uniform or not) over a volume, surface, or line, the amounts of charge dq in a small volume element dV, surface element dA, or length element ds respectively are

$$dq = \rho dV, \qquad dq = \sigma dA, \quad \text{or} \qquad dq = \lambda ds.$$

The electric fields produced by line, surface and volume charge charge distributions respectively are

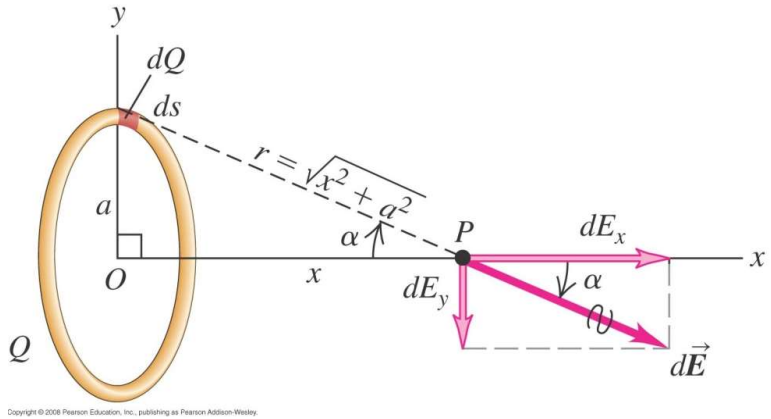
$$\vec{E} = k_e \int \frac{\lambda ds}{r^2}\hat{r}; \qquad \vec{E} = k_e \int \frac{\sigma dA}{r^2}\hat{r}; \qquad \vec{E} = k_e \int \frac{\rho dV}{r^2}\hat{r}.$$

Electric field of a ring of uniformly distributed charge at a point on the axis

- $\lambda = \frac{Q}{l} = \frac{Q}{2\pi a} = \frac{dQ}{ds}$

  $dQ = \lambda ds$

- $dE = k\frac{dQ}{r^2} = k\frac{dQ}{x^2 + a^2}$



- $dE_x = dE\cos\alpha = (k\frac{dQ}{x^2 + a^2}) \times \frac{x}{\sqrt{x^2 + a^2}}$

- $dE_x = dE\cos\alpha = k\frac{\lambda ds}{x^2 + a^2} \times \frac{x}{\sqrt{x^2 + a^2}}$

$$E_x = \int dE_x = \int dE\cos\alpha = \int_0^{2\pi a} k\frac{\lambda ds}{x^2 + a^2} \times \frac{x}{\sqrt{x^2 + a^2}}$$

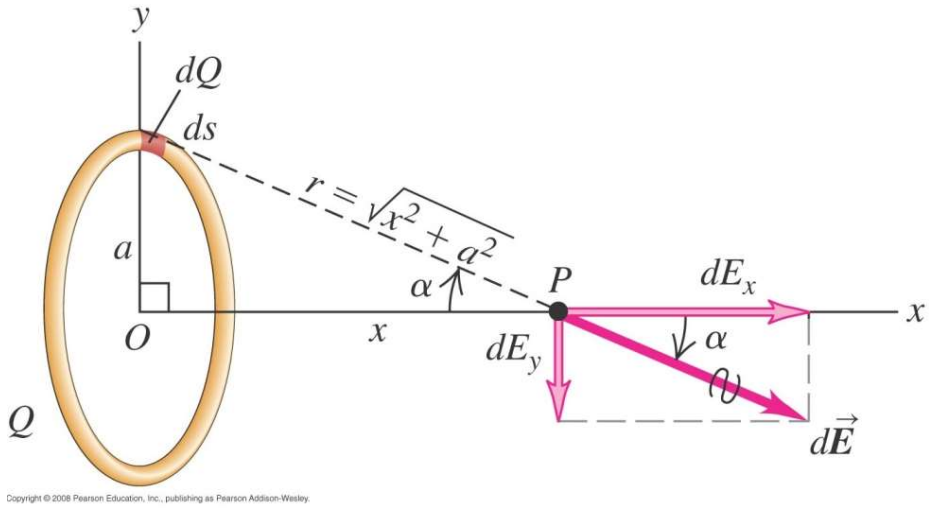
$$E_x = k\frac{\lambda}{x^2+a^2} \times \frac{x}{\sqrt{x^2+a^2}} \int_0^{2\pi a} ds = k\frac{\lambda x}{(x^2+a^2)^{3/2}} \times 2\pi a = k\frac{\lambda 2\pi a x}{(x^2+a^2)^{3/2}} = k\frac{Qx}{(x^2+a^2)^{3/2}}$$

From symmetry, $E_y = 0$.

$$\boldsymbol{E} = Ex\boldsymbol{i} = k\frac{Qx}{(x^2+a^2)^{3/2}}\boldsymbol{i} = \frac{1}{4\pi\varepsilon_0}\frac{Qx}{(x^2+a^2)^{3/2}}\boldsymbol{i}$$

Electric field of a ring of uniformly distributed charge at a point on the axis

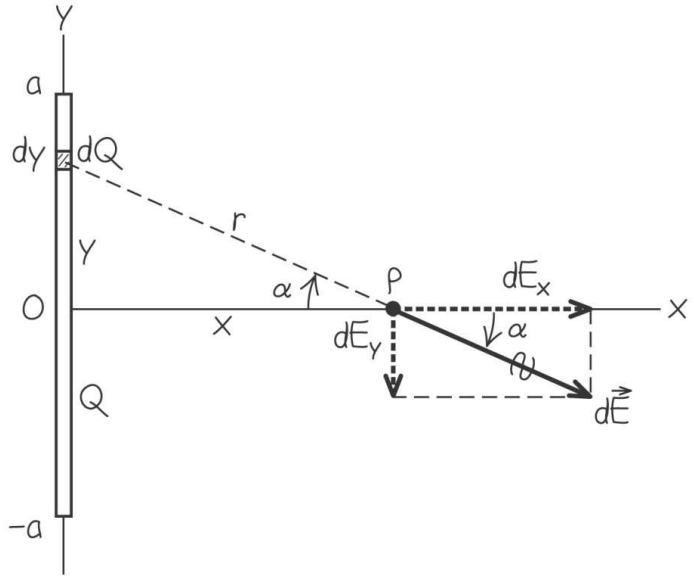
- $\boldsymbol{E} = Ex\boldsymbol{i} = k\frac{Qx}{(x^2+a^2)^{3/2}}\boldsymbol{i} = \frac{1}{4\pi\varepsilon_0}\frac{Qx}{(x^2+a^2)^{3/2}}\boldsymbol{i}$



- At $x >> a$, $\boldsymbol{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{x^2}\boldsymbol{i}$, the electric field is same as a point charge.

- At the centre, x = 0. This implies $\boldsymbol{E}$ = 0 at the centre.

Electric field due to a uniformly charged line segment



- The linear charge density is $\lambda = Q/2a$

$$dQ = \lambda\, dy = (Q/2a)dy$$

$$dE = k\frac{dQ}{r^2} = k\frac{Qdy}{2a(x^2+y^2)}$$

- $dE_x = dE\cos\alpha$
  $dE_y = -\,\mathrm{d}E\,\sin\alpha$

- $$dE_x = k\frac{Qxdy}{2a(x^2+y^2)^{3/2}} \qquad dE_y = -k\frac{Qydy}{2a(x^2+y^2)^{3/2}}$$

$$E_x = \int_{-a}^{a} k\frac{Qxdy}{2a(x^2+y^2)^{3/2}} = k\frac{Q}{x(x^2+a^2)^{1/2}}$$

From symmetry $E_y = 0$

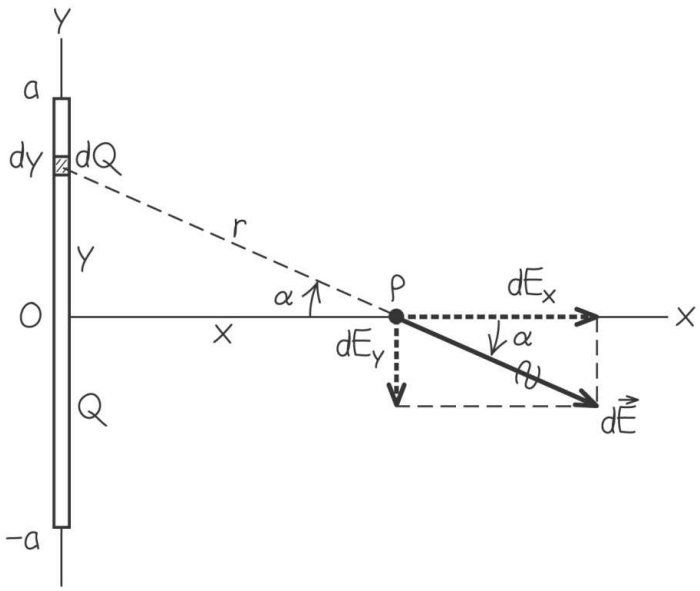
You can also try to evaluate $\int_{-a}^{a} k\frac{Qydy}{2a(x^2+y^2)^{3/2}}$ and you should get zero.

Therefore, $\vec{\boldsymbol{E}} = \vec{E}_x = \frac{kQ}{x(x^2+a^2)^{1/2}}\hat{\imath} = \frac{k2a\lambda}{x(x^2+a^2)^{\frac{1}{2}}}\hat{\imath}$

Electric field due to a uniformly charged line segment

- $\vec{E} = \vec{E}_x = \frac{kQ}{x(x^2+a^2)^{1/2}}\hat{\imath} = \frac{k2a\lambda}{x(x^2+a^2)^{\frac{1}{2}}}\hat{\imath}$

$$\vec{E} = \frac{1}{2\pi\epsilon_0}\frac{\lambda}{x\sqrt{(x^2/a^2)+1}}\hat{\imath}$$

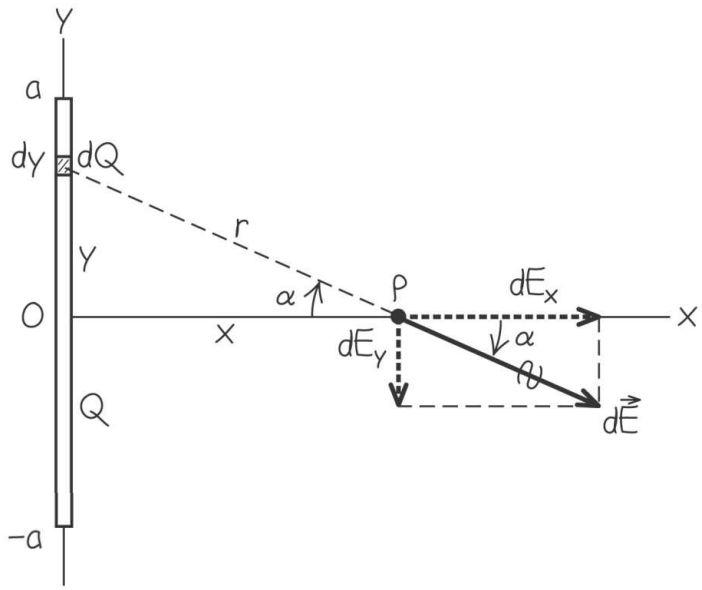


- If $x >> a$, i.e, if the segment is very short or if the field point is very far from the line segment similar to a point charge,

$$\boldsymbol{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{x^2}\boldsymbol{i}$$

Electric field due to a uniformly charged line segment

If $a>>x$, i.e, if the segment is very long, we



so that

$$\vec{E} = \frac{1}{2\pi\epsilon_0}\frac{\lambda}{x\sqrt{(x^2/a^2)+1}}\hat{i}$$

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 x}\hat{i}$$

This is the field of an *infinitely long* line of charge. At any point P at a perpendicular distance r from the line in *any* direction, $\vec{E}$ has magnitude

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \qquad \text{(infinite line of charge)}$$

Note that this field is proportional to $1/r$ rather than to $1/r^2$ as for a point charge.

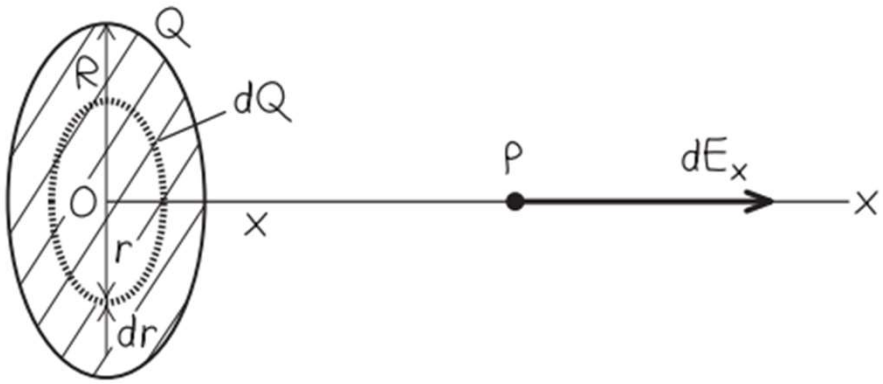
Electric field of a uniformly charged disk along the axis

- $\sigma = \frac{Q}{A} = \frac{dQ}{dA}$

$$dA = 2\pi r\, dr$$

- $$dQ = \sigma\, dA = 2\pi\sigma r\, dr$$

$$dE_x = \frac{1}{4\pi\epsilon_0}\frac{2\pi\sigma r x\, dr}{(x^2 + r^2)^{3/2}}$$



- $$E_x = \int dE_x = \int_0^R \frac{1}{4\pi\epsilon_0}\frac{(2\pi\sigma r\, dr)x}{(x^2 + r^2)^{3/2}} = \frac{\sigma x}{4\epsilon_0}\int_0^R \frac{2r\, dr}{(x^2 + r^2)^{3/2}}$$

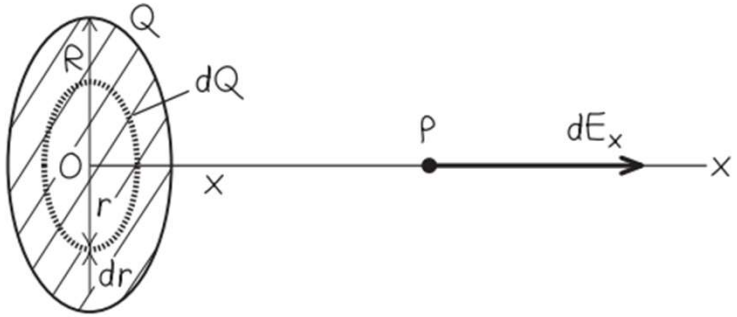
$$E_x = \frac{\sigma x}{2\epsilon_0}\left[-\frac{1}{\sqrt{x^2 + R^2}} + \frac{1}{x}\right]$$

$$= \frac{\sigma}{2\epsilon_0}\left[1 - \frac{1}{\sqrt{(R^2/x^2) + 1}}\right]$$

Electric field of a uniformly charged disk along the axis

$$E_x = \frac{\sigma x}{2\epsilon_0}\left[-\frac{1}{\sqrt{x^2+R^2}}+\frac{1}{x}\right]$$

$$= \frac{\sigma}{2\epsilon_0}\left[1-\frac{1}{\sqrt{(R^2/x^2)+1}}\right]$$



- If $R >> x$, i.e, if the disk is very large or if the point is very close to it

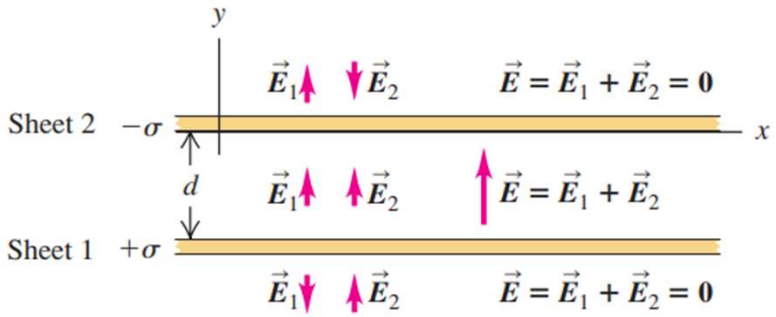
$$E = \frac{\sigma}{2\epsilon_0}$$

Our final result does not contain the distance x from the plane. Hence the electric field produced by an *infinite* plane sheet of charge is *independent of the distance from the sheet*. The field direction is everywhere perpendicular to the sheet, away from it.

If P is to the *left* of the plane ($x < 0$), the result is the same except that the direction of $\vec{E}$ is to the left instead of the right. If the surface charge density is negative, the directions of the fields on both sides of the plane are toward it rather than away from it.

Example

Two infinite plane sheets with uniform surface charge densities $+\sigma$ and $-\sigma$ are placed parallel to each other with separation d as shown. Find the electric field between the sheets, above the upper sheet, and below the lower sheet.



$$E_1 = E_2 = \frac{\sigma}{2\epsilon_0}$$

Between the sheets, $\vec{E}_1$ and $\vec{E}_2$ reinforce each other; above the upper sheet and below the lower sheet, they cancel each other. Thus the total field is

$$\vec{E} = \vec{E}_1 + \vec{E}_2 = \begin{cases} \mathbf{0} & \text{above the upper sheet} \\ \dfrac{\sigma}{\epsilon_0}\hat{j} & \text{between the sheets} \\ \mathbf{0} & \text{below the lower sheet} \end{cases}$$

Electric dipoles

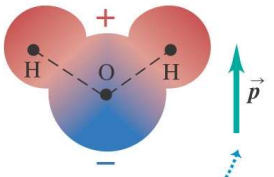
- An electric dipole is an arrangement of a pair of charges that are equal in magnitude but opposite sign separated by a certain distance.

The product of the charge q and the separation d is the magnitude of a quantity called the **electric dipole moment,** denoted by p:

$$p = qd \qquad \text{(magnitude of electric dipole moment)}$$

- Electric Dipole moment is a vector along the dipole axis with the direction from negative to positive charge
- Water molecule is an example of a dipole.

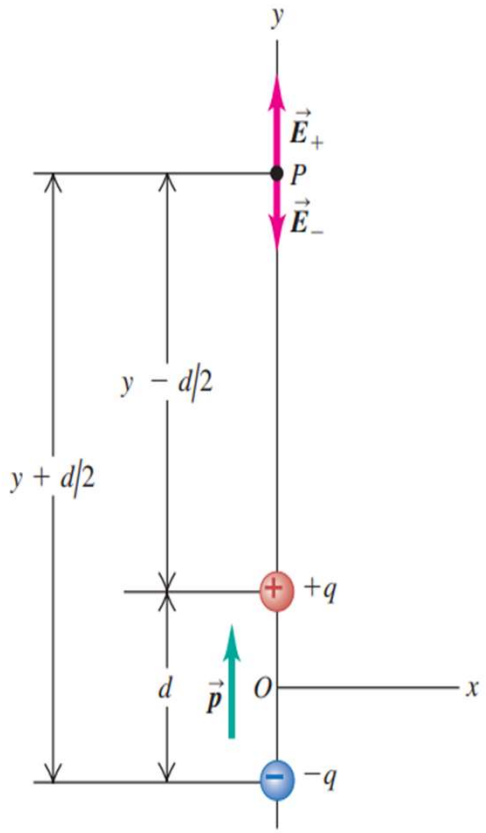
(a) A water molecule, showing positive charge as red and negative charge as blue



The electric dipole moment $\vec{p}$ is directed from the negative end to the positive end of the molecule.

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Electric field due to electric dipole at a point along the *y*-axis



- The total field at P is along *y*-axis which is

$$E_y = \frac{q}{4\pi\epsilon_0}\left[\frac{1}{(y - d/2)^2} - \frac{1}{(y + d/2)^2}\right]$$

$$= \frac{q}{4\pi\epsilon_0 y^2}\left[\left(1 - \frac{d}{2y}\right)^{-2} - \left(1 + \frac{d}{2y}\right)^{-2}\right]$$

Electric field due to electric dipole at a point along the *y*-axis

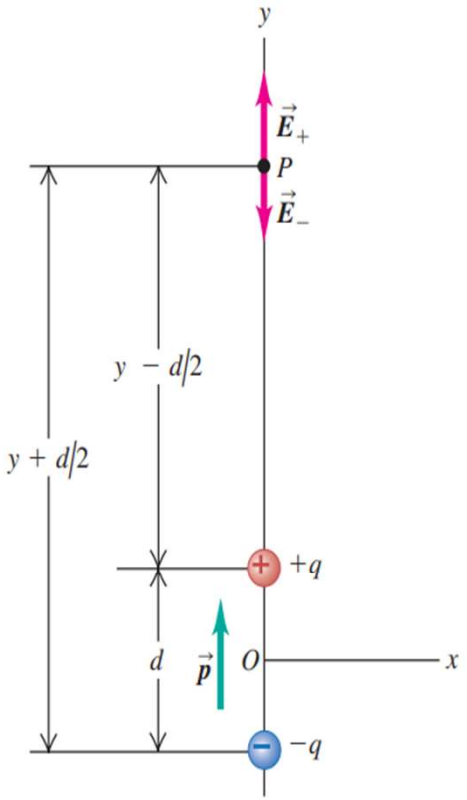
- Using Binomial expansion

$(1 + x)^n \cong 1 + nx + n(n-1)x^2/2 + \cdots$ (valid for the case $|x| < 1$).

- For $y >> d$, $d/2y << 1$

n = -2 and $d/2y$ replaces x in the Binomial expansion so that

$$\left(1 - \frac{d}{2y}\right)^{-2} \cong 1 + \frac{d}{y} \quad \text{and} \quad \left(1 + \frac{d}{2y}\right)^{-2} \cong 1 - \frac{d}{y}$$

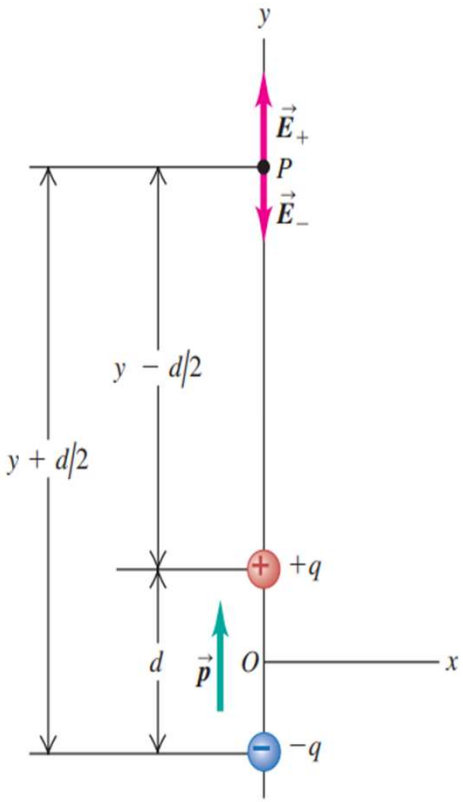


Hence E_y is given approximately by

$$E_y \cong \frac{q}{4\pi\epsilon_0 y^2}\left[1 + \frac{d}{y} - \left(1 - \frac{d}{y}\right)\right] = \frac{qd}{2\pi\epsilon_0 y^3} = \frac{p}{2\pi\epsilon_0 y^3}$$

Electric field due to electric dipole at a point along the *y*-axis

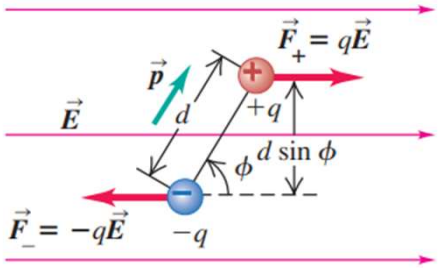
$$E_y \cong \frac{q}{4\pi\epsilon_0 y^2}\left[1 + \frac{d}{y} - \left(1 - \frac{d}{y}\right)\right] = \frac{qd}{2\pi\epsilon_0 y^3} = \frac{p}{2\pi\epsilon_0 y^3}$$



- For large distances away from dipole, the field drops as $1/r^3$ which is different from a point charge ($1/r^2$) and long line charge ($1/r$) and large sheet of charge (which is independent of r).

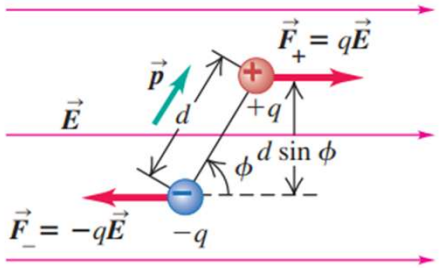
Force and Torque on an electric dipole in a uniform electric field

- The net force on an electric dipole is zero.



- There will be an external torque, $\boldsymbol{\tau}$. With respect to the centre, each force will contribute a torque of magnitude qE $(d/2)$ $\sin\phi$.
- The total magnitude $\tau = qEd \sin\phi = pE \sin\phi$
- The direction of torque is into the page.

Torque on an electric dipole in a uniform electric field



- Recalling the cross product of vectors, $\vec{\tau} = \vec{p} \times \vec{E}$

- Torque is maximum when ***p*** and ***E*** are perpendicular to each other and zero when they are parallel (stable equilibrium) or antiparallel (unstable equilibrium).

The torque will try to turn ***p*** to line it up with ***E***.

Potential Energy of an Electric dipole

- Recalling an infinitesimal work done by the torque is

$$dW = \tau \, d\phi$$

- We know that the torque due to the electric field will act in the direction of decreasing angle so we first write $\tau = -pE \sin\phi$

Therefore, $dW = -pE \sin\phi d\phi$

$$W = \int dW = \int_{\phi_1}^{\phi_2} -pE \sin\phi \, d\phi = pE \cos\phi_2 - pE \cos\phi_1$$

Potential Energy (U) of an Electric dipole

We know that $W = -\Delta U = -(U_2 - U_1) = U_1 - U_2$

$$U_1 - U_2 = pE\cos\phi_2 - pE\cos\phi_1$$
$$= -pE\cos\phi_1 + pE\cos\phi_2$$
$$= -pE\cos\phi_1 - (-pE\cos\phi_2)$$

We therefore write $U = -pE\cos\phi = -\boldsymbol{p}.\boldsymbol{E}$

The potential energy has its minimum (most negative) value $U = -pE$ at the stable equilibrium position, where $\phi = 0$ and $\vec{\boldsymbol{p}}$ is parallel to $\vec{\boldsymbol{E}}$. The potential energy is maximum when $\phi = \pi$ and $\vec{\boldsymbol{p}}$ is antiparallel to $\vec{\boldsymbol{E}}$; then $U = +pE$. At $\phi = \pi/2$, where $\vec{\boldsymbol{p}}$ is perpendicular to $\vec{\boldsymbol{E}}$, U is zero.

End of chapter